DATABASE SCHEMA DOCUMENTATION

Web-Based ML Model Evaluator

This document describes the database schema implemented for the Web-Based ML Model Evaluator. The backend uses Flask-SQLAlchemy with SQLite as the default database, configurable through the DATABASE_URL environment variable.

1. Database Overview

The schema contains five tables: users, datasets, models, evaluation_results, and user_sessions. Together they store accounts, dataset metadata, model records, results, and revocable login sessions.

- ORM: Flask-SQLAlchemy / SQLAlchemy
- Default database: SQLite
- Default database file: backend/ml_evaluator.db
- Initialization: init_db(app) registers the models and calls db.create_all().

Note: db.create_all() creates missing tables; it does not migrate existing tables when model definitions change.

2. users

Purpose: Stores user account information.

Column	Data Type	Constraints	Description
id	INTEGER	Primary key, NOT NULL	Unique user identifier.
username	VARCHAR(80)	UNIQUE, NOT NULL	Username.
email	VARCHAR(120)	UNIQUE, NOT NULL	Email address.
password_hash	VARCHAR(256)	NOT NULL	Hashed password value.
created_at	DATETIME	NOT NULL; server timestamp default	Record creation time.

3. datasets

Purpose: Stores dataset metadata and file location.

Column	Data Type	Constraints	Description
id	INTEGER	Primary key, NOT NULL	Unique dataset identifier.
user_id	INTEGER	FK to users.id, NOT NULL	Dataset owner.
filename	VARCHAR(255)	NOT NULL	Recorded filename.
file_path	VARCHAR(512)	NOT NULL	Path to the dataset file.
file_size	BIGINT	NULL allowed	File size, if recorded.
row_count	INTEGER	NULL allowed	Number of rows, if recorded.
column_count	INTEGER	NULL allowed	Number of columns, if recorded.
metadata_json	JSON	NULL allowed	Optional dataset metadata.
created_at	DATETIME	NOT NULL; server timestamp default	Record creation time.

Foreign keys: user_id → users.id

4. models

Purpose: Stores ML model and training configuration metadata.

Column	Data Type	Constraints	Description
id	INTEGER	Primary key, NOT NULL	Unique model-record identifier.
user_id	INTEGER	FK to users.id, NOT NULL	Associated user.
dataset_id	INTEGER	FK to datasets.id, NOT NULL	Dataset used by the model.
model_name	VARCHAR(128)	NOT NULL	Model name.

algorithm	VARCHAR(128)	NOT NULL	Selected algorithm.
model_path	VARCHAR(512)	NULL allowed	Optional saved-model path.
status	VARCHAR(32)	NOT NULL; default 'pending'	Model/training status.
config_json	JSON	NULL allowed	Optional model configuration.
created_at	DATETIME	NOT NULL; server timestamp default	Record creation time.

Foreign keys: user_id → users.id; dataset_id → datasets.id

5. evaluation_results

Purpose: Stores evaluation metrics and optional detailed outputs.

Column	Data Type	Constraints	Description
id	INTEGER	Primary key, NOT NULL	Unique evaluation identifier.
user_id	INTEGER	FK to users.id, NOT NULL	Associated user.
model_id	INTEGER	FK to models.id, NOT NULL	Evaluated model.
metrics_json	JSON	NOT NULL	Evaluation metrics.
confusion_matrix_json	JSON	NULL allowed	Optional confusion matrix.
report_json	JSON	NULL allowed	Optional detailed report.
created_at	DATETIME	NOT NULL; server timestamp default	Record creation time.

Foreign keys: user_id → users.id; model_id → models.id

6. Relationships

Parent table	Child table	Relationship
users	datasets	One user can own multiple datasets.
users	models	One user can have multiple model records.
datasets	models	One dataset can be associated with multiple model records.
users	evaluation_results	One user can be associated with multiple evaluation results.
models	evaluation_results	One model can have multiple evaluation results.

The schema includes user_id on models and evaluation_results. Application logic should ensure that users are authorized to access the datasets, models, and evaluations referenced by those records.

7. Entity Relationship Diagram (Textual)

```
USERS (1) ————————< DATASETS
USERS (1) ————————< MODELS >———————— (1) DATASETS
USERS (1) ————————< EVALUATION_RESULTS >———————— (1) MODELS
```

8. Implementation and Verification

The models are defined in backend/models/user.py, backend/models/dataset.py, backend/models/model.py, and backend/models/evaluation.py. Initialization is implemented in backend/database/db.py, which imports the models before calling db.create_all().

Verification: five tables including user_sessions registered; schema and key relationships inspected.
Automated tests cover revoked-cookie replay, 24-hour idle expiry, and owner-scoped data access.
CSV upload and model creation remain separate planned stories.

9. Future Considerations

- Test create, read, update, and delete operations for each model.

- Test foreign-key behavior and validation of invalid references.
- Verify authorization so users cannot access another user's records.
- Use database migrations if the schema changes after deployment.
- Ensure password values are hashed and plaintext passwords are never stored.

10. Source Files

- backend/config.py
- backend/database/db.py
- backend/models/__init__.py
- backend/models/user.py
- backend/models/dataset.py
- backend/models/model.py
- backend/models/evaluation.py
- backend/models/user_session.py

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11. user_sessions

Purpose: Stores revocable authenticated-session state. The browser cookie contains a signed opaque session ID; this record is authoritative for access, revocation, and idle expiry.

Column	Data Type	Constraints	Description
session_id	VARCHAR(64)	Primary key, NOT NULL	Random per-login identifier.
user_id	INTEGER	FK users.id, NOT NULL	Session owner.
created_at	DATETIME	NOT NULL; server timestamp default	Session issue time.
last_seen_at	DATETIME	NOT NULL	Authenticated activity; idle-expiry clock.
revoked_at	DATETIME	NULL allowed	Logout or expiry revocation time.

Foreign key: user_id -> users.id. One user can have multiple session records.

12. Session lifetime and logout

A successful login creates a random session ID and its server-side record. Protected requests validate the user, record, and revocation state, then refresh last_seen_at. Sessions expire after 24 hours without authenticated activity. Logout marks the record revoked and clears the cookie; replaying a previously captured cookie is rejected even if its signature remains valid.

13. US-07 owner-scoped access

Dataset reads filter by datasets.user_id. Model reads require both models.user_id and the associated datasets.user_id to match the authenticated user. Evaluation reads require the result, model, and dataset owner IDs to match. Missing and foreign IDs return the same not-found response. Filesystem paths are omitted from API representations. Any future artifact download must apply the same model ownership check before opening a saved file.

14. Scope and verification

The CSV upload endpoint (US-08) and model/evaluation creation routes (US-18/US-19) remain planned work. Their write paths must set owner IDs from the authenticated session, never from request data. Automated owner-isolation fixtures seed two users and their datasets, models, and results. See backend/tests/test_auth_logout.py and backend/tests/test_us07_data_isolation.py for case mappings.